Brief Announcement: Delay-Optimal Transaction Order Fairness

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Abstract

Order-fair consensus aims to prevent a leader or block producer from exploiting transaction order, a concern amplified by front-running and MEV in decentralized finance. Existing order-fairness notions avoid some impossibilities by batching cyclic dependencies or by allowing bounded displacement, but these relaxations do not distinguish a tiny timing inversion from a large physical time gap. We revisit approximate-order-fairness (AOF), originally introduced and dismissed as too weak or impossible in prior work, in a synchronous model where parties can timestamp transaction arrivals using physical time. We show that time-aware AOF has a nontrivial feasible region: if a fraction φ of nodes receive tx at least θ before tx′, then tx can be forced before tx′ whenever $\theta > \Delta_{\mathsf{sync}}/k$ and $\varphi > 1 - h/k$, where h is the honest fraction and Δ_{sync} is the honest dissemination bound. We also give matching-style infeasibility constructions showing why smaller delays or lower thresholds permit Condorcet cycles. Finally, we outline how a HotStuff-style consensus layer can agree on timestamp reports while a deterministic ordering function enforces the strongest acyclic AOF constraints available in the observed execution.

CCS Concepts

• **Theory of computation → Distributed algorithms**; • **Security and privacy → Distributed systems security**.

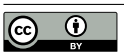
Keywords

Byzantine consensus, order fairness, transaction ordering, MEV

ACM Reference Format:
Zhuo Cai and Amir Goharshady. 2026. Brief Announcement: Delay-Optimal Transaction Order Fairness. In *ACM Symposium on Principles of Distributed Computing (PODC '26), July 6–10, 2026, Egham, United Kingdom.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3796701.3815912

1 Introduction

Order-fair consensus was proposed to limit the ordering power of Byzantine leaders in blockchain protocols [10], with follow-up work improving performance, latency, or the target guarantee [2, 9, 11, 12, 14, 16]. This line is motivated by front-running and MEV in decentralized finance [6, 8, 13, 15, 17, 18, 21]: if an attacker sees a victim transaction early enough to react, then the final ledger should not let the attacker's later transaction execute first.



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PODC '26, Egham, United Kingdom
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ACM ISBN 979-8-4007-2512-8/26/07
https://doi.org/10.1145/3796701.3815912

The difficulty is that pairwise receive orders can form Condorcet cycles. Prior definitions therefore relax strict receive-order fairness by batching mutually cyclic transactions, by bounding how far a transaction may be displaced, or by using very strong timed conditions. These relaxations are useful, but they treat all majority edges similarly. A majority edge supported by a one-microsecond gap and a majority edge supported by a ten-second gap are both just edges. In financial settings this loses the quantity that matters most: physical time.

We revisit approximate-order-fairness (AOF), which requires ordering only when enough nodes saw a sufficiently large time gap. AOF was already present in earlier work [10], but was largely set aside because of reductions from strict receive-order fairness in round-based models. Our starting point is that modern blockchain systems can use physical-time observations: honest nodes cannot be made to confuse a padded logical execution with a real execution in which transactions arrived far apart in wall-clock time. This brief announcement summarizes the feasibility landscape; the full version is available at HAL hal-05625798 [3].

Contributions. This brief announcement makes three contributions. First, we show that vote-only fairness notions can lose a unanimously large physical-time gap once auxiliary transactions create weak Condorcet-style cycles. Second, we map nontrivial feasible and infeasible AOF parameter regimes in a synchronous network, relating the required delay θ, support threshold φ, honest fraction h, and dissemination bound Δ_{sync}. Third, we sketch a validation-oriented protocol: consensus agrees on signed timestamp reports, and validators reconstruct the AOF graph and check the canonical topological order.

2 Model and Definition

There are n nodes $\mathcal{S}$, with honest set $\mathcal{H}$ of size hn and Byzantine set $\mathcal{A}$. We assume a synchronous dissemination bound Δ_{sync}: if one honest node receives a valid transaction, every honest node receives it within Δ_{sync}. Node i records a timestamp $\mathsf{ts}_i(\mathsf{tx})$ when it receives tx; Byzantine nodes may report arbitrary timestamps.

For two transactions tx and tx′, define

$$\mathcal{I}_\theta(\mathsf{tx}, \mathsf{tx}') = \{i \in \mathcal{S} : \mathsf{ts}_i(\mathsf{tx}) + \theta \le \mathsf{ts}_i(\mathsf{tx}')\}.$$

We write $\mathsf{tx} \prec_{\theta,\varphi} \mathsf{tx}'$ if $|\mathcal{I}_\theta(\mathsf{tx}, \mathsf{tx}')| \ge \varphi n$. A ledger satisfies (θ, φ)-AOF if every such relation is respected by the final ledger order. The useful range is $1 - h < \varphi \le h$: below it, Byzantine nodes alone can impose arbitrary constraints; above it, even unanimous honest evidence is insufficient.

AOF is implementable exactly when its constraint graph is acyclic: then deterministic topological sorting satisfies all constraints, while a directed cycle makes the required ledger order contradictory. This graph view separates the semantic question of whether chosen parameters can generate contradictory obligations from the protocol

question of whether honest validators compute the same graph. The complete synchrony model and formal fairness definitions are in the full version [3, Sec. 2].

3 Why Time Magnitudes Matter

Our first observation is that fairness definitions based only on pairwise voting can fail to distinguish weak local inversions from strong global time gaps.

Theorem 3.1 (Timed-cycle weakness). *For any threshold $\varphi \in (1/2, 1)$ and any time gap $D > 0$, there is a Δ_{sync}-synchronous arrival profile in which every node receives tx_A at least D before tx_B, yet auxiliary transactions form a pairwise φ-majority cycle from tx_B back to tx_A.*

Proof idea. Choose $k > 1/(1-\varphi)$ and a small $\delta < \Delta_{\mathsf{sync}}/k$. For each transition in a chain, a φ fraction of nodes observe a forward gap δ, while the remaining nodes observe a large backward jump inside the allowed Δ_{sync} window. After k such transitions the support pattern can be reset while the entire honest receive window has shifted left by roughly $\Delta_{\mathsf{sync}} - k\delta > 0$. Repeating the construction shifts the window far enough left to close a cycle from a transaction after tx_B back to tx_A, despite the unanimous large gap tx_A before tx_B. □

Thus a definition that ignores time magnitudes may legally batch or fail to force the order between tx_A and tx_B once the adversary supplies enough chaff transactions. AOF filters out the weak edges in this construction while preserving the large-gap edge. The construction is a definitional stress test: it does not force a bad implementation choice, but shows that the definition lacks a strong enough obligation to protect the large-gap pair once weak auxiliary edges close a cycle. This is the front-running-relevant case: a reactive transaction may be separated from the victim by real time, yet auxiliary transactions with small timing differences can dilute the formal guarantee. The full construction and implications are in the full version [3, Secs. 3.1–3.3].

4 Relation to Existing Guarantees

The distinction is clearest when comparing what each fairness notion is allowed to forget. Block-order fairness, as used by Aequitas [10] and related protocols, handles cycles by allowing transactions to share a batch or block. This avoids contradictory pairwise obligations, but it leaves the intra-block order outside the guarantee. AOF instead asks which pairwise obligations survive after small timing gaps are discarded. When the resulting graph is acyclic, the guarantee is strict for those surviving pairs.

Quick Order Fairness [2] is orthogonal: it relaxes vote support for speed, while AOF changes what a report may support by requiring a physical-time gap. Median-timestamp, blind-ordering, and separation-based approaches address aggregation, privacy, and efficiency on other axes [4, 14, 20].

Timed-order fairness, as in Wendy [12], uses a stronger timing condition: all honest nodes must receive one transaction before any honest node receives the other. In our notation this corresponds to a coarse separation larger than the honest receive window. AOF keeps that regime but also covers overlapping receive windows when enough reports show a large enough delay.

Bounded-unfairness [11] measures how many ledger positions a transaction can be displaced. This is useful, but position distance is not reaction time: the former counts unrelated traffic, while the latter measures the opportunity to observe and respond. In high-throughput DeFi, many unrelated transactions may separate victim and reaction, while the attacker only needs the physical time gap to respond. AOF therefore uses time as the primitive quantity and identifies when the position error can be reduced to zero for a supported pair; see the full comparison and case study in [3, Secs. 6.1–6.5].

5 Feasibility Landscape

The next theorem gives the main positive region; complete proofs and trade-off analysis are in the full version [3, Sec. 4].

Theorem 5.1 (Feasible AOF region). *In a Δ_{sync}-synchronous network with honest fraction h, fix a positive integer k such that $1 - h/k \le h$. If*

$$\theta > \Delta_{\mathsf{sync}}/k \qquad \text{and} \qquad \varphi > 1 - h/k,$$

then the (θ, φ)-AOF graph is acyclic for every execution profile. Hence AOF can be achieved by deterministic topological sorting.

Proof idea. Assume, for contradiction, a cycle $\mathsf{tx}_1 \prec \mathsf{tx}_2 \prec \cdots \prec \mathsf{tx}_\ell \prec \mathsf{tx}_1$. Let s_j be the earliest honest receive time of tx_j. For each edge, fewer than an h/k fraction of all nodes, hence fewer than a $1/k$ fraction of honest nodes, may fail to support the edge. Starting from the earliest honest receipt of tx_j, after one edge more than a $(k-1)/k$ fraction of honest nodes must receive the next transaction after time $s_j + \Delta_{\mathsf{sync}}/k$; after two edges more than a $(k-2)/k$ fraction must be shifted by another Δ_{sync}/k; and so on. After k edges, the earliest honest receive window has moved strictly right, giving $s_j < s_{j+k}$. Applying this around the whole cycle and summing the strict inequalities yields $\sum_j s_j < \sum_j s_j$, a contradiction. □

The boundary is meaningful. For $\theta \le \Delta_{\mathsf{sync}}/k$ and $\varphi < 1 - h/(k+1)$, one can partition honest nodes into $k+1$ groups and rotate their receive times across a Δ_{sync} window so that each consecutive edge in a transaction cycle is supported by a $k/(k+1)$ fraction of honest nodes. Byzantine nodes can be partitioned into finer groups to raise support arbitrarily close to $1 - h/(k+1)$. This produces a genuine AOF cycle, so no ledger can satisfy all constraints.

Theorem 5.2 (Boundary cycles). *In the same model, for every positive integer k with $1 - h/k \le h$, parameters $\theta \le \Delta_{\mathsf{sync}}/k$ and $\varphi < 1 - h/(k+1)$ admit a valid synchronous profile whose AOF graph contains a directed cycle.*

Proof idea. Partition honest nodes into $k+1$ equal groups and place their timestamps for transactions $\mathsf{tx}_0, \mathsf{tx}_1, \ldots$ at offsets $0, \Delta_{\mathsf{sync}}/k, \ldots, \Delta_{\mathsf{sync}}$ in a rotating pattern. For every consecutive pair, exactly k of the $k+1$ honest groups observe the forward delay Δ_{sync}/k. Byzantine nodes are split into many smaller rotating groups so that, for each edge, all but one Byzantine group also supports the forward direction. As the number of Byzantine groups grows, the support approaches $(k/(k+1))h + (1-h) = 1 - h/(k+1)$. Choosing enough transactions closes the rotation into a directed cycle, contradicting any ledger order. □

The gap between $1 - h/k$ and $1 - h/(k+1)$ reflects a real granularity trade-off. Large delays, such as $\theta > \Delta_{\text{sync}}$, require only a low threshold above the Byzantine fraction. Finer delays, such as $\theta \approx \Delta_{\text{sync}}/k$, require stronger near-unanimous support.

When h is close to $2/3$, the coarse regime $k = 1$ enforces gaps larger than Δ_{sync} with any threshold above the Byzantine fraction. The next regime $k = 2$ enforces gaps just above $\Delta_{\text{sync}}/2$, but only with a threshold above roughly $2/3$. Thus AOF does not promise arbitrary timestamp precision for free: finer timing requires evidence from a larger fraction of the system, and unavailable evidence should leave the pair unconstrained.

6 Protocol Sketch

The feasibility result can be carried by a standard BFT consensus protocol; the full version gives the deterministic ordering function and carrier protocols in detail [3, Secs. 5.1–5.3]. Each node signs a timestamp report for the transactions it has observed. A leader proposes a block containing a report set $\mathcal{R}$, and the consensus protocol agrees on that report set together with the transaction list. Validators check:

(1) every report is correctly signed and bound to the current slot;
(2) the report set satisfies the quorum or synchrony-specific inclusion rule;
(3) the transaction order equals the deterministic output of $\text{Order}(\mathcal{R})$.

The ordering function computes, for each pair $(\text{tx}_a, \text{tx}_b)$, the largest delay supported by at least $\lceil \varphi n \rceil$ reports, adds the corresponding AOF edge when the target delay is met, and outputs a canonical topological sort, breaking ties by transaction hash. Equivalently, a delay-optimal version can lower the target delay until the graph is just acyclic, thereby enforcing the strongest acyclic constraints visible in the agreed reports.

This validation rule is local: validators verify signatures, reconstruct pairwise delays and the graph, and check the canonical output, rather than infer the leader's intention or trust a claimed certificate. If the leader changes a timestamp, omits required evidence, or chooses a different tie-breaking order, the block fails validity. The fairness argument is therefore reduced to the inclusion properties of $\mathcal{R}$.

In a fully synchronous deployment, one can instantiate the carrier with Sync HotStuff [1]. If a leader omits a node's report, it must include either evidence of equivocation or a quorum of absence signatures collected after the report deadline. Since an honest report broadcast on time reaches all honest validators within Δ_{sync}, such a certificate cannot be formed against an honest sender. Thus all honest reports are preserved, and the feasibility theorem applies with the original honest fraction h.

The same timing discipline also gives a bounded-dependency rule for block production. Suppose a slot collects reports up to a deadline. For coarse parameters, a transaction reported sufficiently before the deadline brings with it all transactions that could θ-precede it; a transaction first seen too late cannot retroactively create a valid large-gap edge into the already selected prefix. This is the operational counterpart of the timed-cycle analysis: by ignoring edges whose physical gaps are below θ, the protocol avoids waiting indefinitely for future chaff transactions that might complete a purely vote-based cycle.

In a partially synchronous HotStuff-style deployment [19], the leader may be allowed to proceed with $n - f$ reports for liveness. The price is a weaker effective honest fraction: in the worst case, the included set contains f Byzantine reports and omits f honest reports, leaving only $n - 2f$ honest reports among $n - f$. The same ordering function still applies, but the feasible threshold must be calibrated to this reduced report set.

Thus the theorem is a feasibility statement for synchronous timestamp evidence, not for every partially synchronous execution. Safety and liveness may still come from the carrier protocol, but fairness during unstable periods must use the weaker effective report set or wait until the timing assumptions for θ hold.

7 Scope and Caveats

The model isolates the ordering layer. It does not by itself solve transaction inclusion, mempool privacy [5], or censorship resistance [7]. A leader that can exclude a transaction can still harm users unless the consensus layer also enforces an inclusion rule or the application has an independent availability mechanism. The role of AOF is narrower: once a transaction pair is represented in the agreed report set, the leader should not be able to reverse a sufficiently well-supported physical-time ordering.

The timing assumption should also be read operationally. If deployed clocks have bounded skew or timestamping uncertainty, that error budget must be absorbed into the chosen delay parameter. The theorem needs honest timestamp evidence that is comparable at the granularity claimed by θ; it does not justify enforcing sub-clock-error orderings. This is why the parameter trade-off is useful: a conservative system can choose coarse delays with lower support thresholds, while a tightly synchronized system can spend higher support to obtain finer ordering.

Finally, the positive and negative regions intentionally leave a small boundary gap. The feasible theorem rules out all AOF cycles above $\varphi > 1 - h/k$ when $\theta > \Delta_{\text{sync}}/k$, while the cyclic construction rules out thresholds below $1 - h/(k+1)$ at $\theta \le \Delta_{\text{sync}}/k$. Closing this gap would require a sharper characterization of how adversarial reports and honest receive-window shifts interact. The important point is that the feasible region is nonempty, parameterized, and compatible with standard BFT validation.

Acknowledgments

This work was partially supported by the ERC Starting Grant 101222524 (SPES). Z. Cai was supported by the Hong Kong PhD Fellowship Scheme. Authors are ordered alphabetically.

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